WCPMS - Web Crop Phenology Metrics Service

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Introduction

Historically, human beings have observed and documented the life cycles of animals and plants, which gave rise to the term Phenology (Demaree, 2009).

Phenology is the part of biology that studies the periodic and cyclical processes of living beings and analyzes how they are influenced by seasonal and interannual variations in climate or habitat (Lieth, 2013).



Source: pexels

Introduction

Land Surface Phenology (LSP) is the term used for vegetation phenology derived from Earth observation (EO) satellite images (Caparros-Santiago, 2021).

LSP usually uses vegetation indices calculated from spectral properties of satellite images for vegetation change detection (Zeng, 2020).

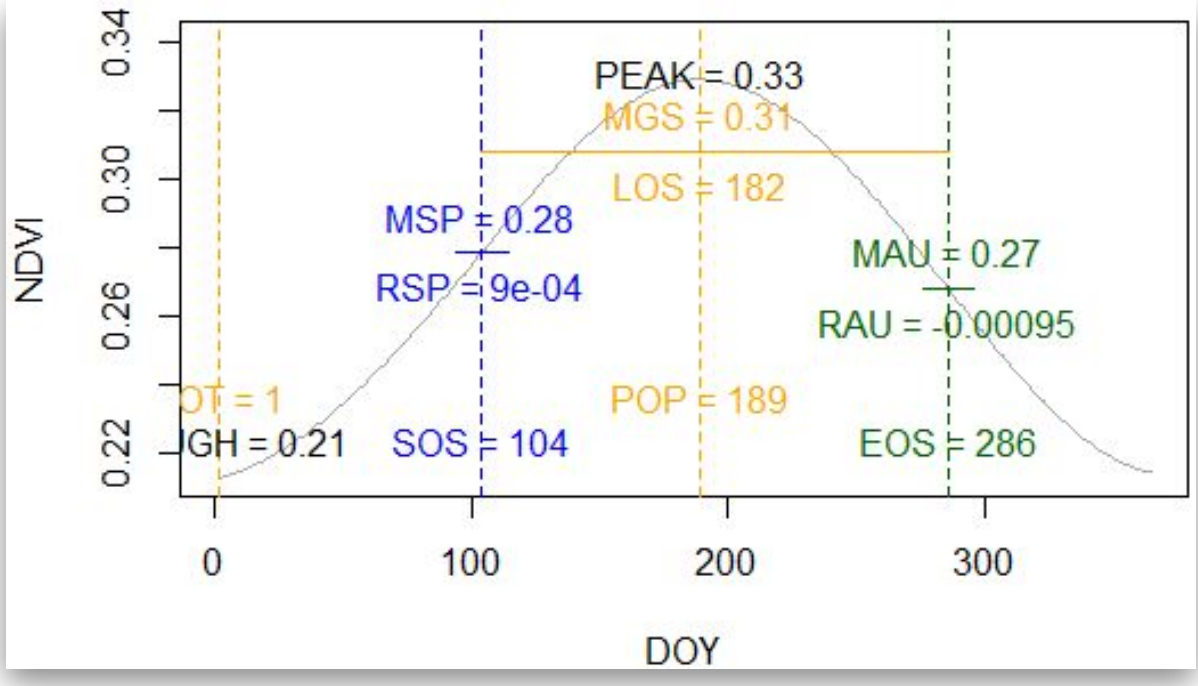


Source: BDC

Introduction

Phenology metrics are a set of measurements associated with satellite image time series that allow the estimation of phenological phases or stages (Rodigheri, 2023).

Some examples of metrics are: start of greening/season (SOS), peak of growing season (POS), onset of senescence or end of the season (EOS), and growing season length (LOS) (Jonsson, 2002).

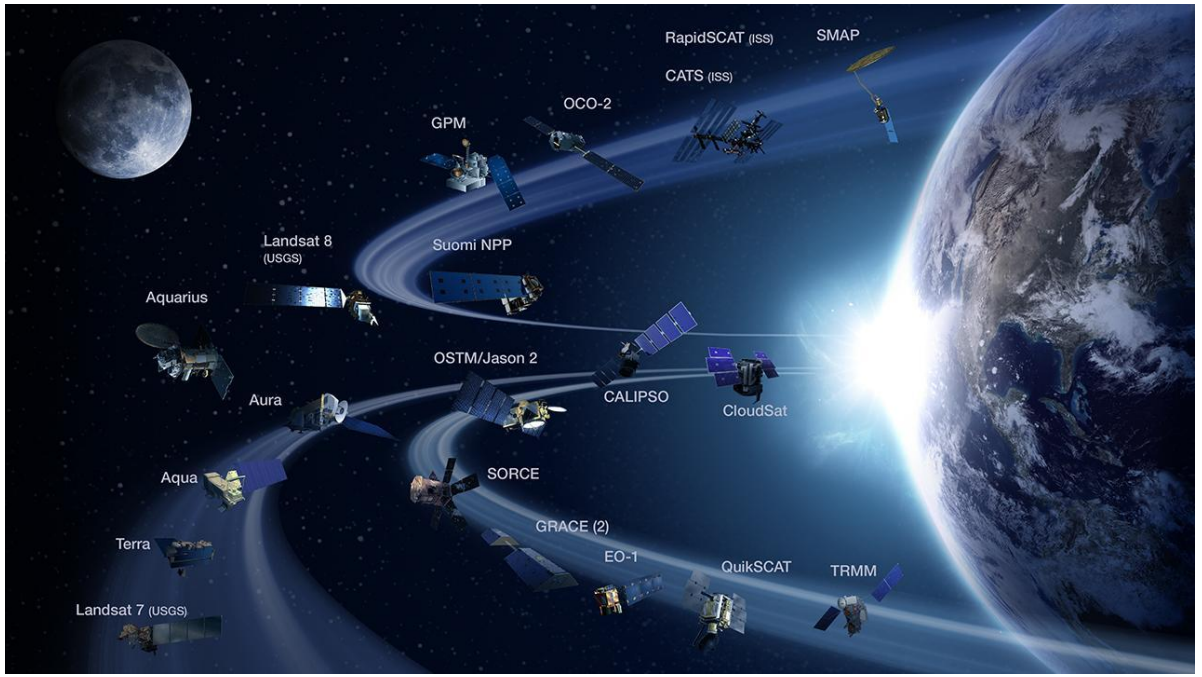


Source: (Forkel M, 2015)

Introduction

Today, large volumes of images are produced by different EO satellites, with distinct spatial, temporal, and spectral resolutions, and made freely available to society.

These large EO data sets pose a challenge for storage and processing systems, which exceeds the capacity of personal computers to handle them.

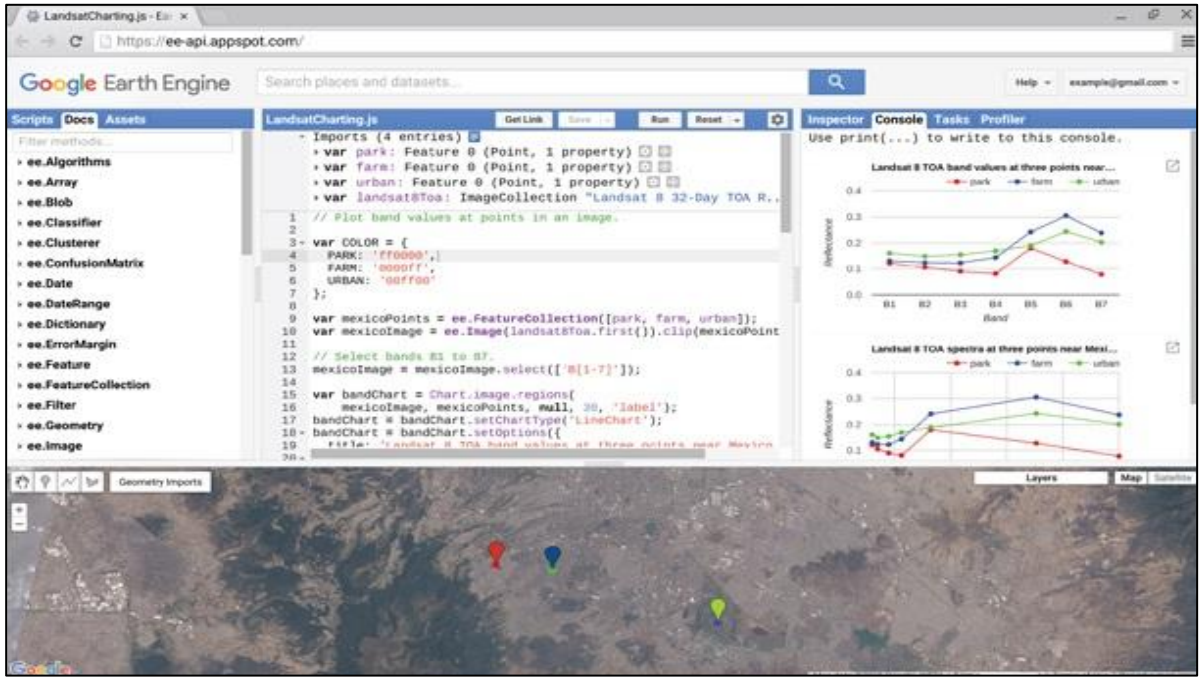


Source: NASA

Introduction

It requires a change of paradigm and a move away from traditional local processing and data distribution methods.

To address this challenge, novel software platforms have been developed to access and handle big amounts of images in cloud computing environments, such as Google Earth Engine (Gomes, 2024).

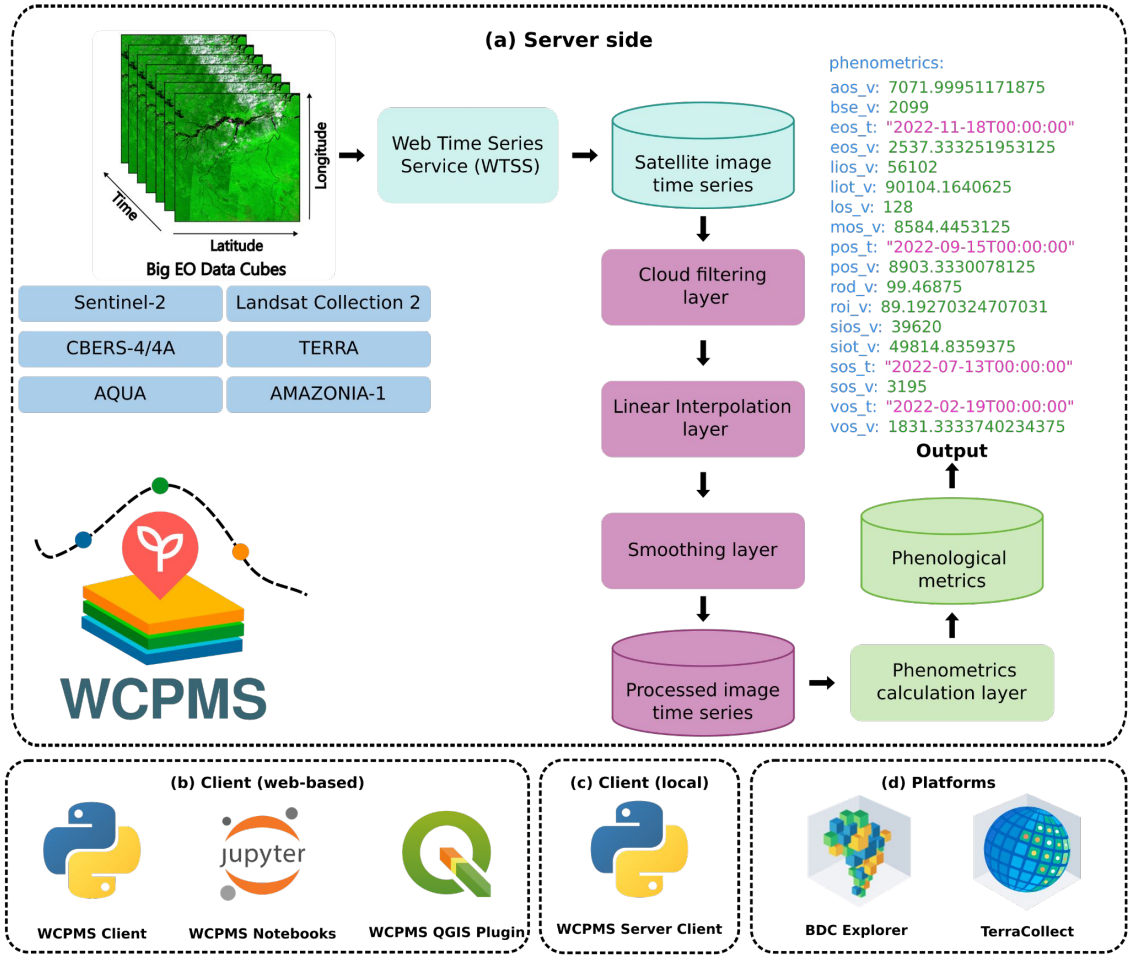


Source: GEE

Introduction

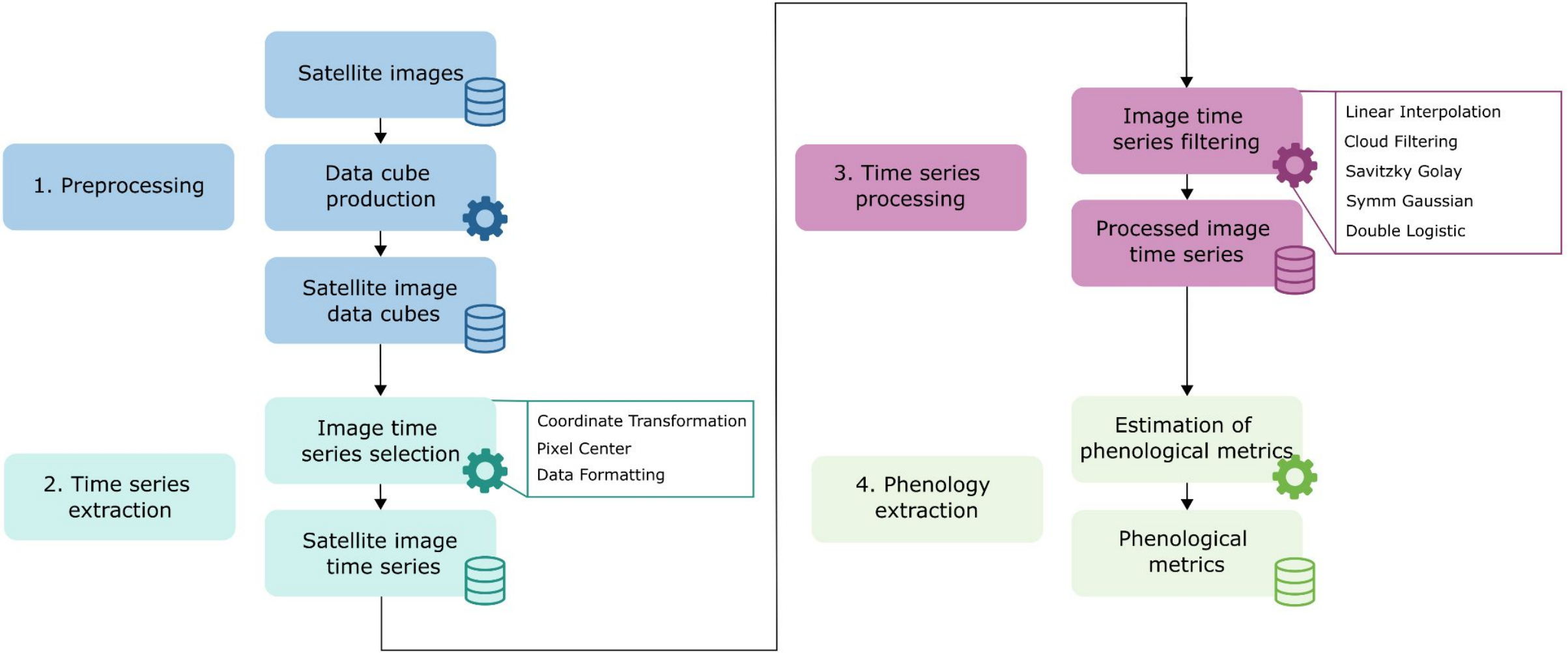
This work presents **a tool for phenology metrics analysis** from big EO image collections that runs in server-side infrastructures, without needing to download large EO data and to install any package on personal computers.

The modules of the proposed tool are: the WCPMS service; multiple clients to access the WCPMS service, such as Python client; a QGIS plugin; and components for the web interfaces, BDCExplorer and TerraCollect.



Source: Author

Crop Phenology workflow

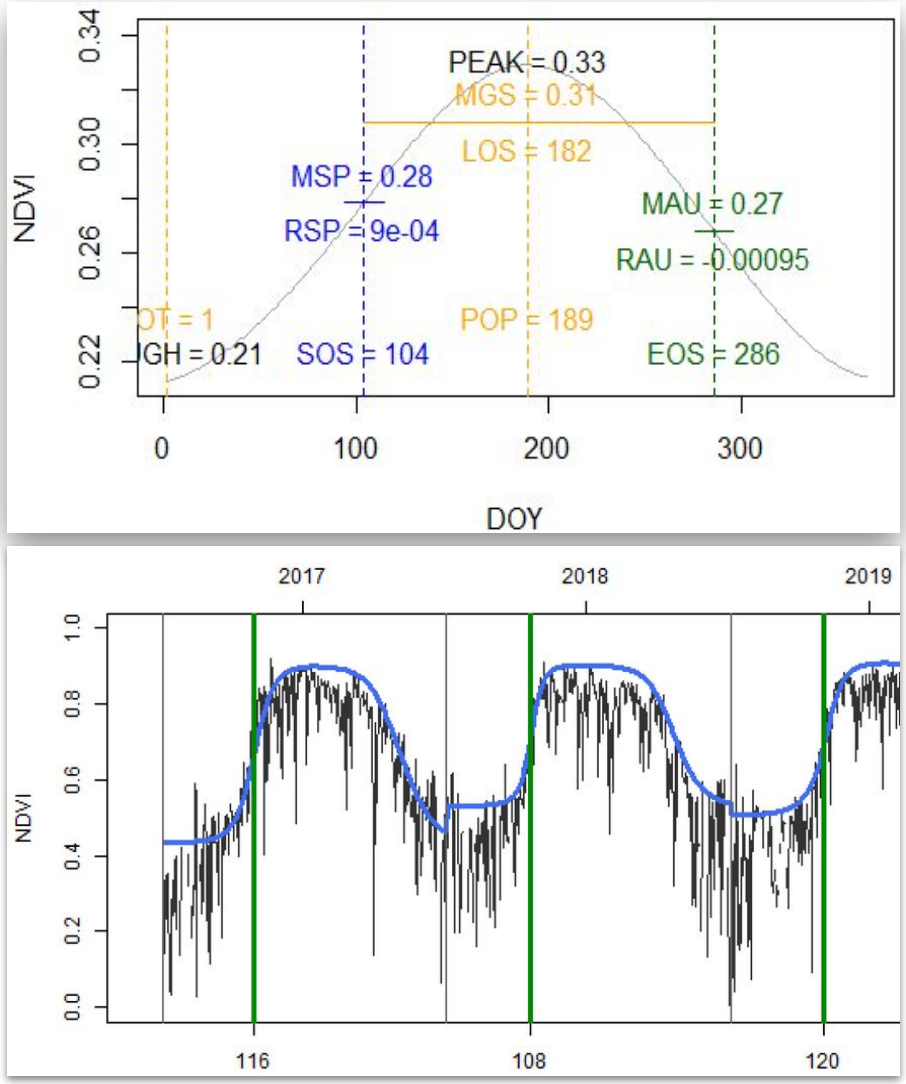


Source: Author

Packages

This section presents a comparative analysis of on-premises tools and packages for crop phenological metrics extraction. Instead of developing novel algorithms, we opted to integrate existing solutions in the proposed tool.

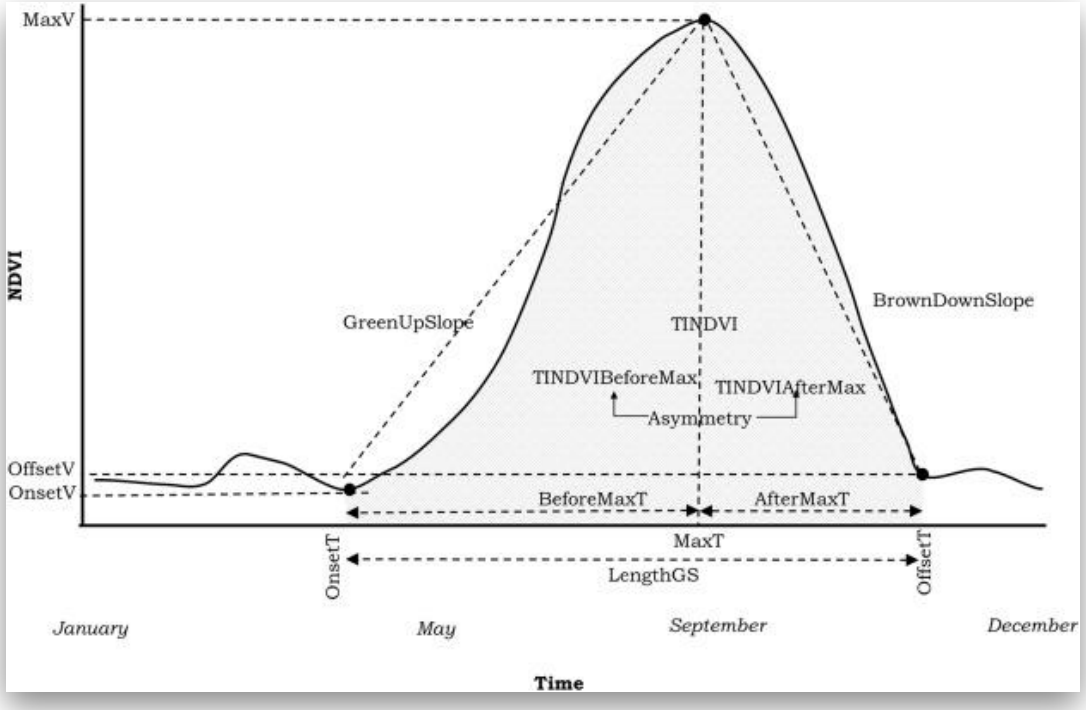
The choice of an existing system was based on this comparative analysis.



Source: (Forkel M, 2015; Şenel, 2023)

CropPhenology

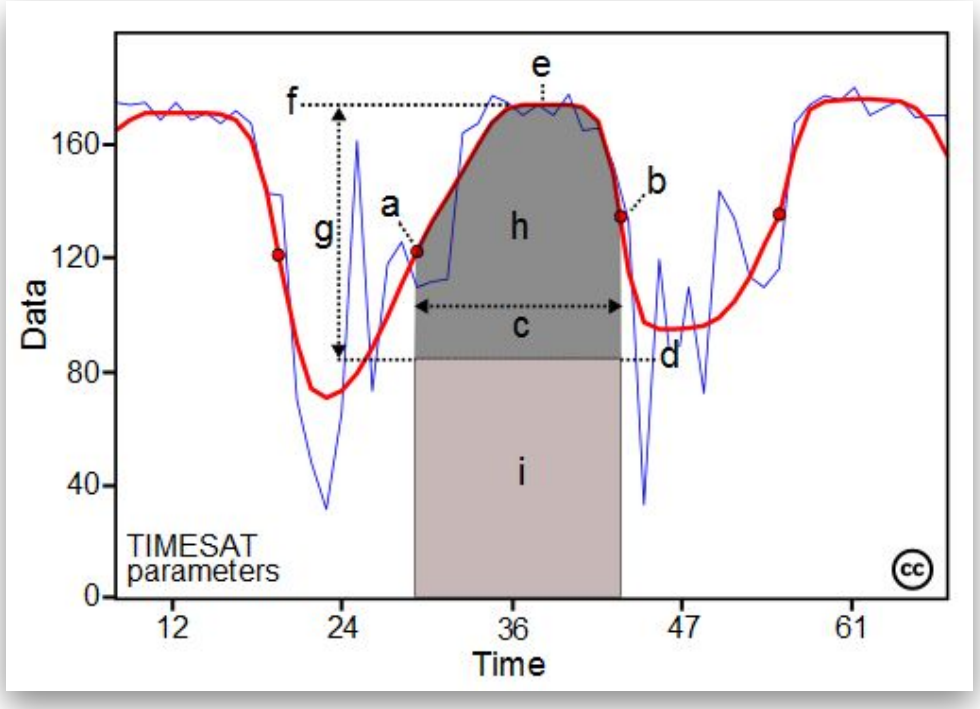
The **CropPhenology** is an R package for extracting crop phenology from time series remotely sensed vegetation index imagery. Developed by Araya, this package allows users to extract and map 15 crop phenological metrics from vegetation index time series of satellite images.



Source: (Araya, 2018)

TIMESAT

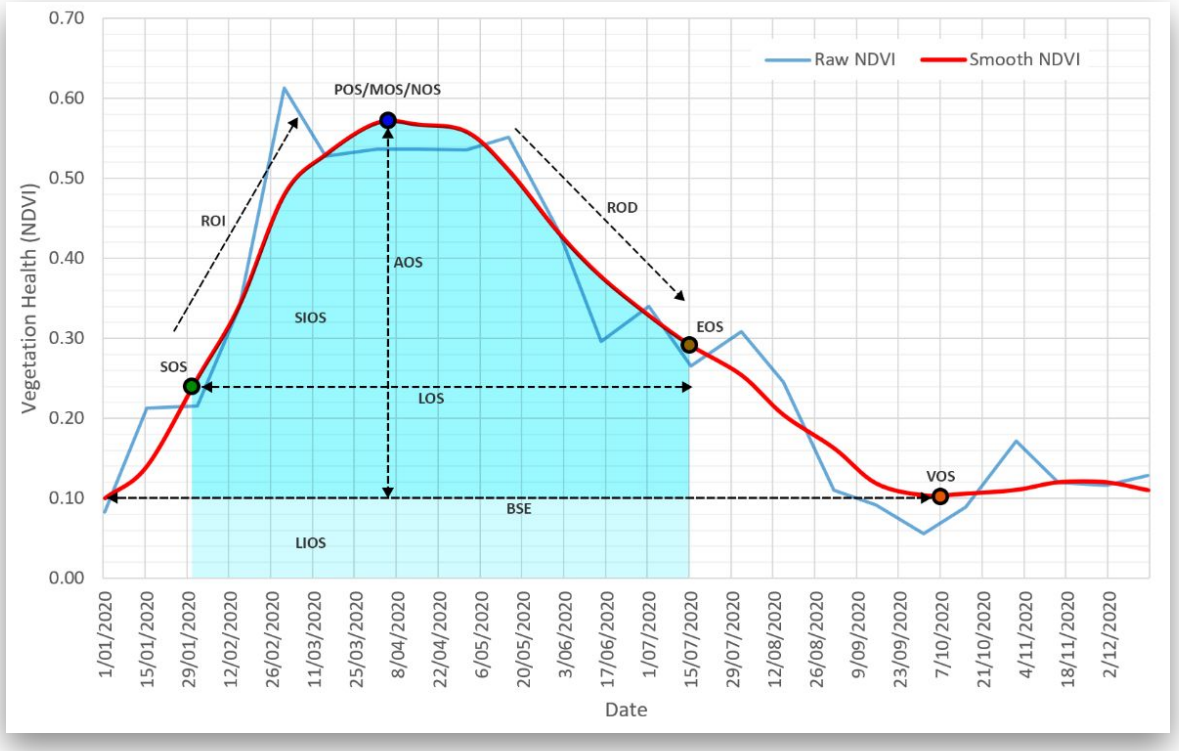
TIMESAT is the most established package for extracting phenological metrics. It is a software package developed in Fortran and Matlab for estimating growing seasons and for computing phenological metrics from satellite image time series. Besides Fortran and Matlab implementations, TIMESAT has implementations in other programming languages, such as R (rTIMESAT) and Python (Phenolopy).



Source: (Jonsson, 2002)

Phenolopy

Phenolopy is a Python package for analyzing satellite image time series. This library is heavily based on the concept and methodology behind the TIMESAT software. A feature of the library is that it can be applied to any satellite imagery stack. The Phenolopy package allows users to extract 19 crop phenological metrics from a given satellite image time series.



Source: PhenoloPy

Software as a service

To align with the SaaS model, we built a distributed architecture. Part of the workflow runs on the server side, and part runs on the client side.

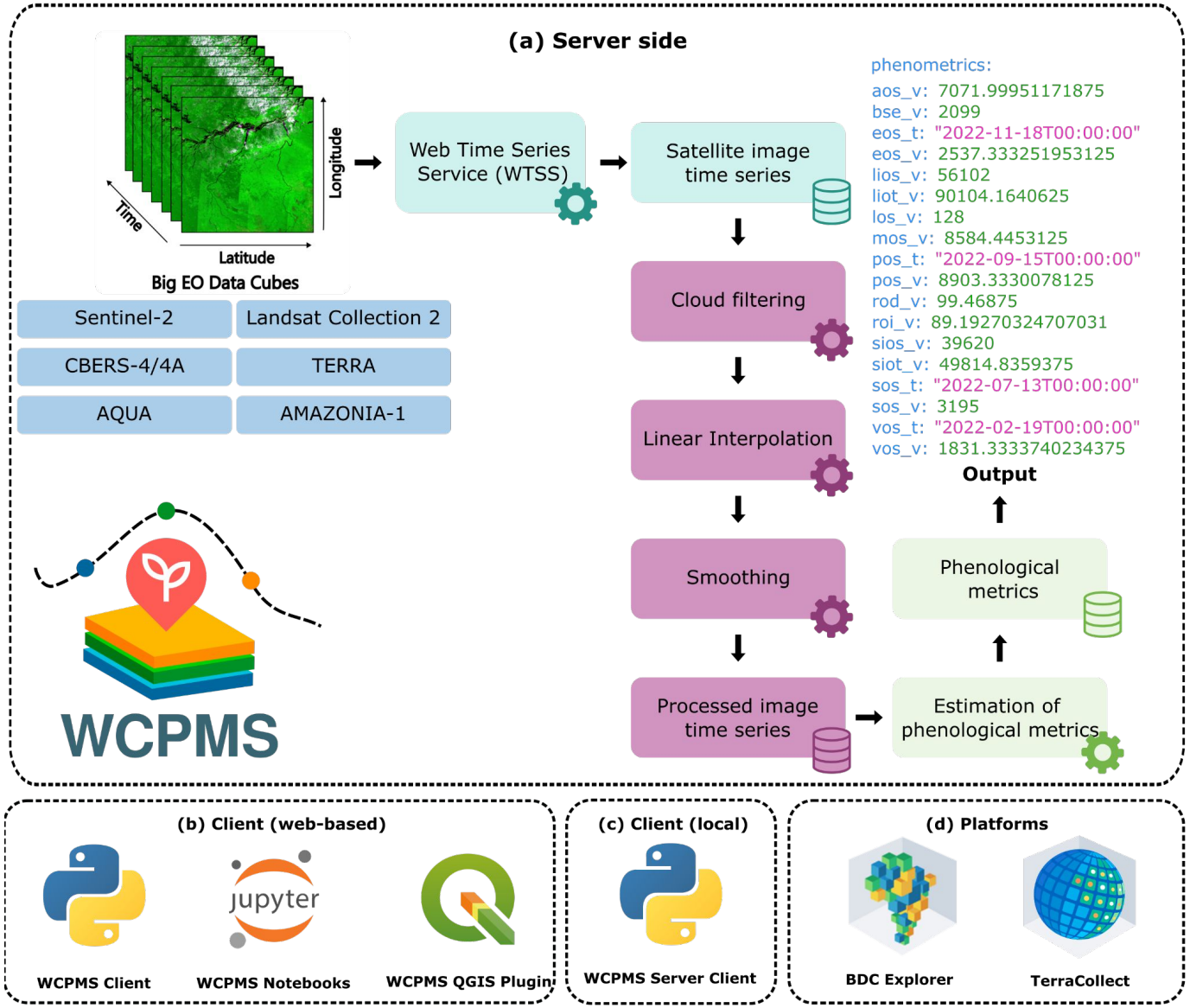
For data access, we integrated the tool with the BDC Web Time Series Service, allowing easy retrieval of satellite image time series.

Guided by Rodigheri (2023), we chose the Phenolopy library for version 1.0. Importantly, our modular design allows for the easy integration.



Source: BDC

A tool for crop phenology metrics analysis



WCPMS

The primary module of the tool is a web service called Web Crop Phenology Metrics Service (WCPMS). This modules was made as a web service, a type of software design to facilitate the communication between different applications over the internet. Through the WCPMS, analysts can give a spatial location and it retrieves the phenological metrics associated to it, using image time series.

#request
https://data.inpe.br/bdc/wcpms/phenometrics?collection=S2-16D-2&latitude=-29.20263&longitude=-55.9554&start_date=2022-01-01&end_date=2022-12-31&freq=16D&band=NDVI

#response


```
{
  "phenometrics": {
        "eos_t": "2022-11-18T00:00:00",
        "eos_v": 2537.333251953125,
        "sos_t": "2022-07-13T00:00:00",
        "sos_v": 3195.0,
        …
  },
  "timeseries": {
        "timeline": [
        "2022-01-01",
        "2022-01-17",
        "2022-02-02",
        ...
  ],
        "values": [
        5593.0,
        2879.0,
        1799.0,
        ...
  ]
```

WCPMS Py Client

A key module of the tool for crop phenology analysis is the Python client. We created the wcpms.py library from scratch to facilitate crop phenology extraction operations. This library was developed to be interoperable with other Python libraries, thus enabling users to integrate established libraries into their own workflows for pre- or post-processing and analysis.

```
from wcpms import *

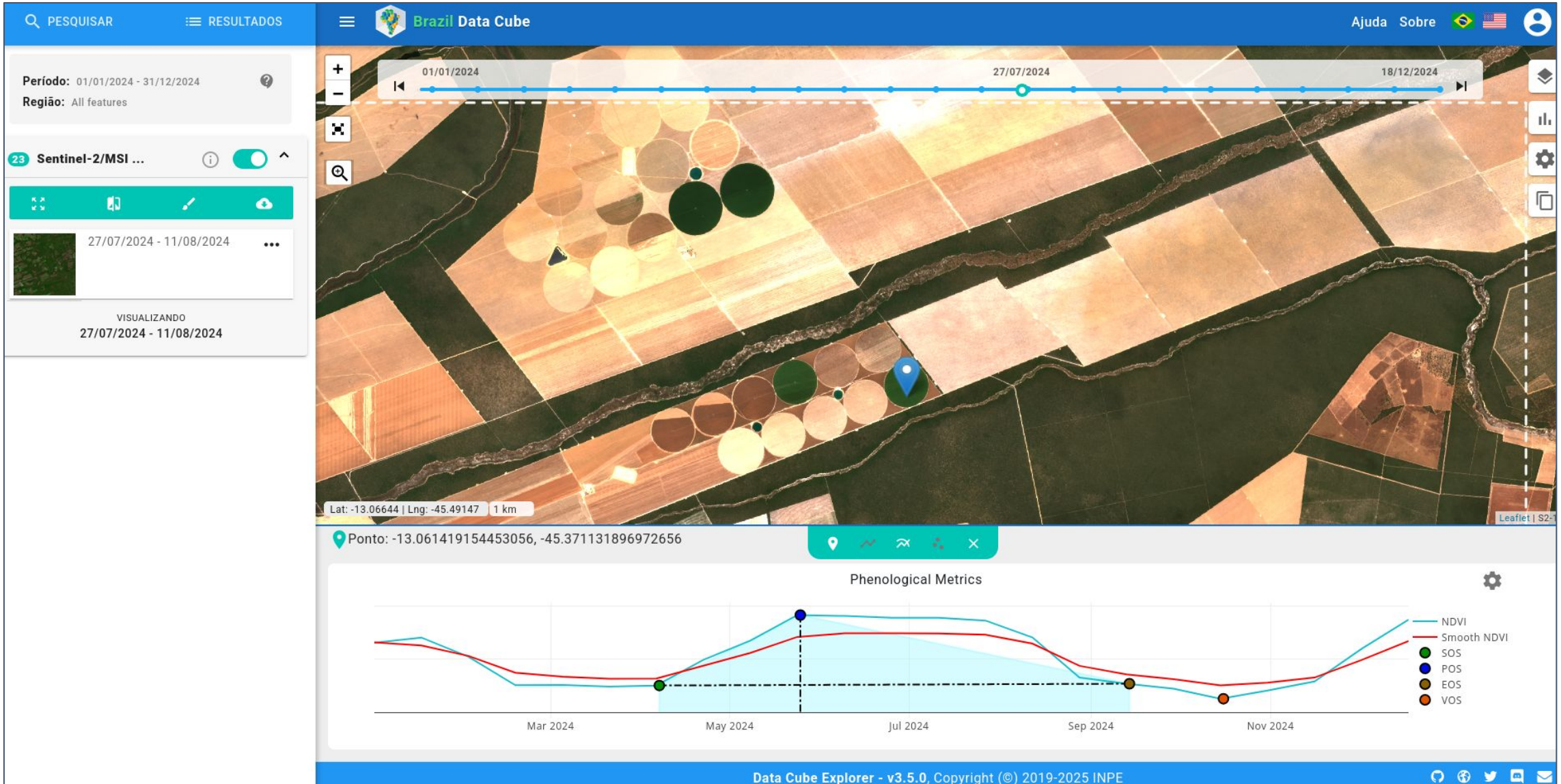
# Defines URL of a instance of the WCPMS
wcpms_url = "http://data.inpe.br/bdc/wcpms"

# Defines a data cube query from Brazil Data Cube (BDC)
datacube = cube_query(
        collection="S2-16D-2",
        start_date="2022-01-01",
        end_date="2022-12-31",
        freq="16D",
        band="NDVI"
)

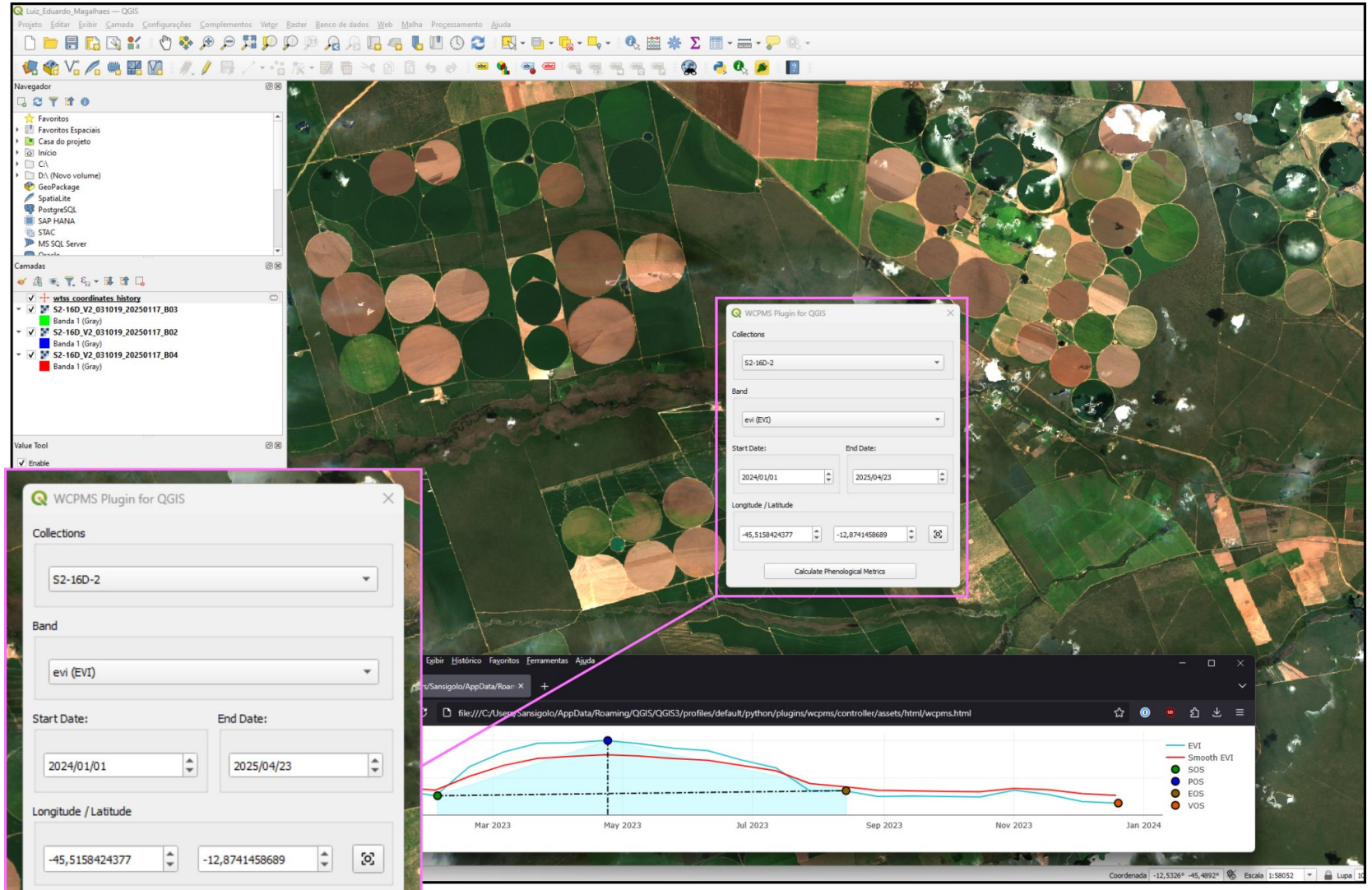
# Retrieving the phenological metrics
pm = get_phenometrics(
        url = wcpms_url,
        cube = datacube,
        latitude=-13.67705998895876,
        longitude=-52.47134616677471
)

# Visualizing the phenological metrics
print(pm)
```

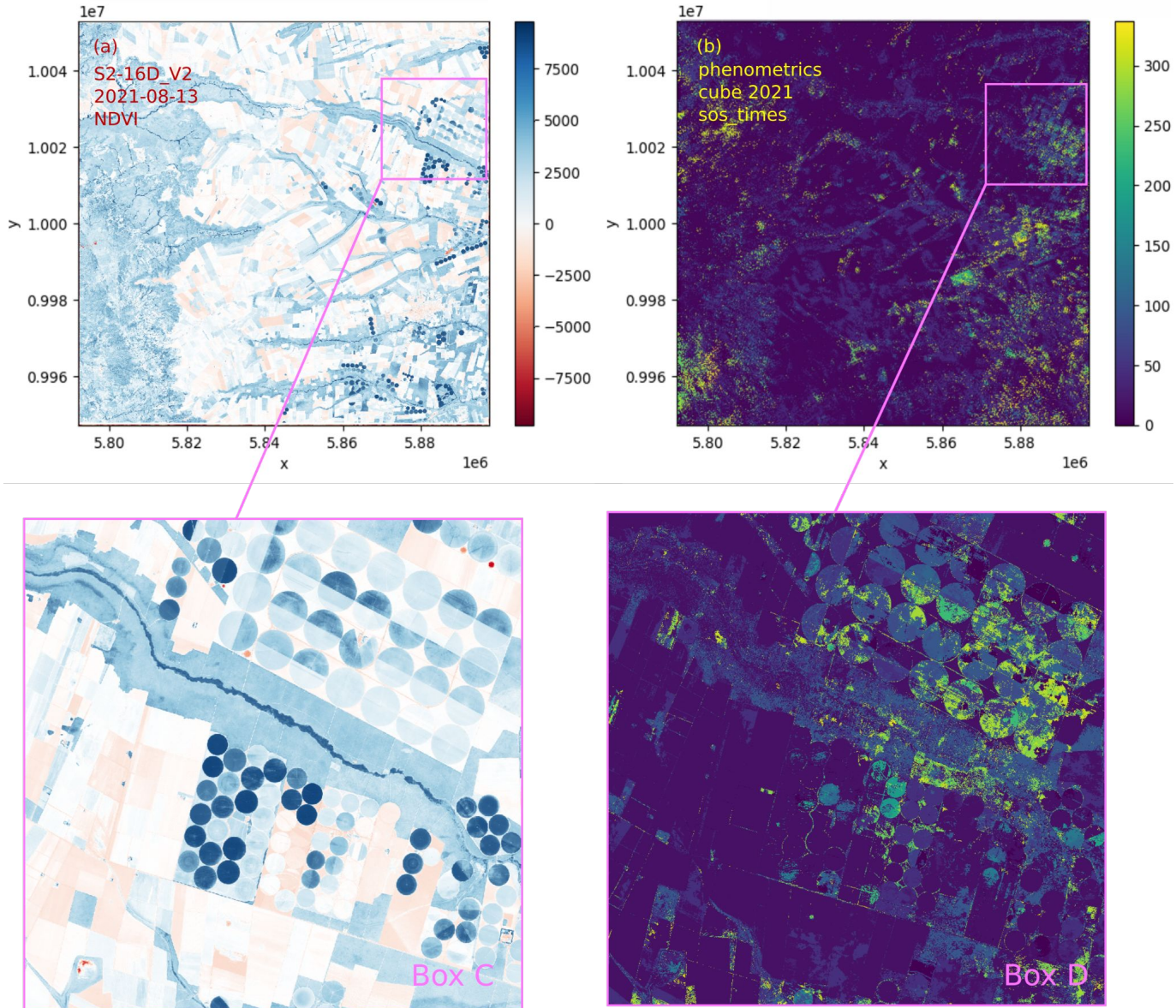
WCPMS BDC Explorer



WCPMS QGIS plugin



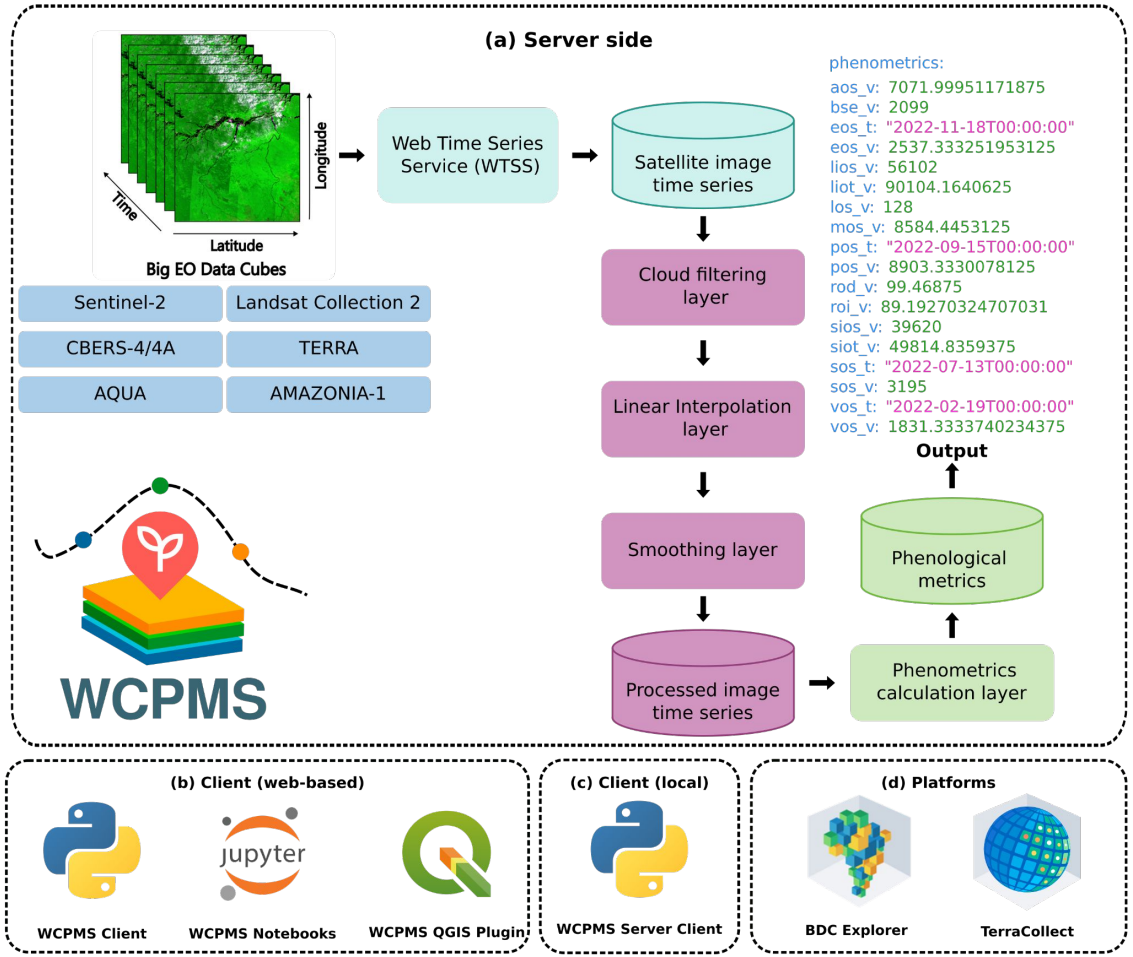
Data cube crop phenology calculation



Conclusions

We managed to create a novel tool for crop phenology metrics analysis from big EO image collections produced by the BDC project.

The tool delivers the functionalities of established algorithms for crop phenology metrics extraction connect to a rich set of data cubes from diverse remote sensing constellations, including Sentinel-2, Landsat Collection 2, CBERS, AMAZONIA and more.



Source: Author

Code availability

PyPI wcpms.py:
https://pypi.org/project/wcpms

Github wcpms_server:
https://github.com/GSansigolo/wcpms

Github Paper Compedium:
https://github.com/GSansigolo/tool-for-crop-phenology-paper

Github wcpms.py:
https://github.com/brazil-data-cube/wcpms.py

Read the Docs WCPMS:
https://wcpms.readthedocs.io/en/latest/

Github WCPMS QGIS:
https://github.com/brazil-data-cube/wcpms-qgis

Github WCPMS Angular:
https://github.com/brazil-data-cube/ng-wcpms

Github WCPMS HTML:
https://github.com/brazil-data-cube/html-wcpms

Thank you!

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